



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

Lecture 19: Additional and Dominant Poles and Zeros

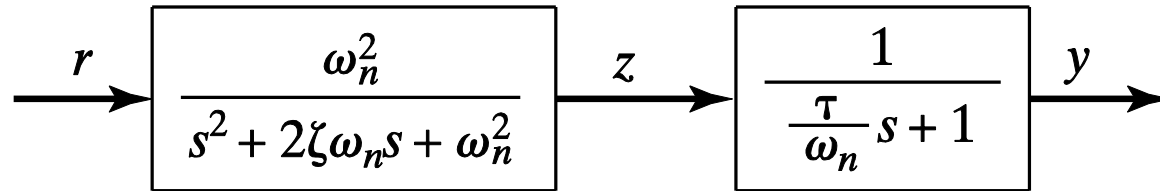
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Additional Pole or Zero

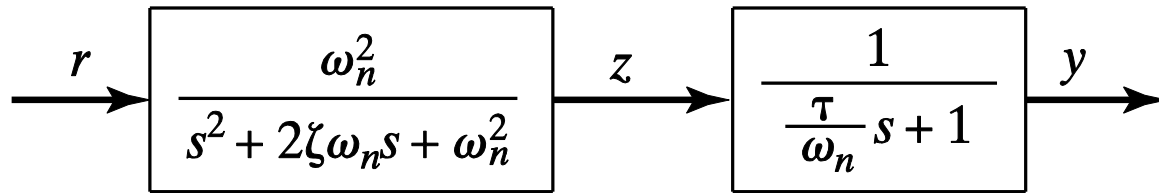
- Most systems can be better modeled using **high-order** transfer functions (i.e., having more than two poles) and have finite zeros.
- Understand how these **additional poles** and **zeros** affect the **standard** system response.



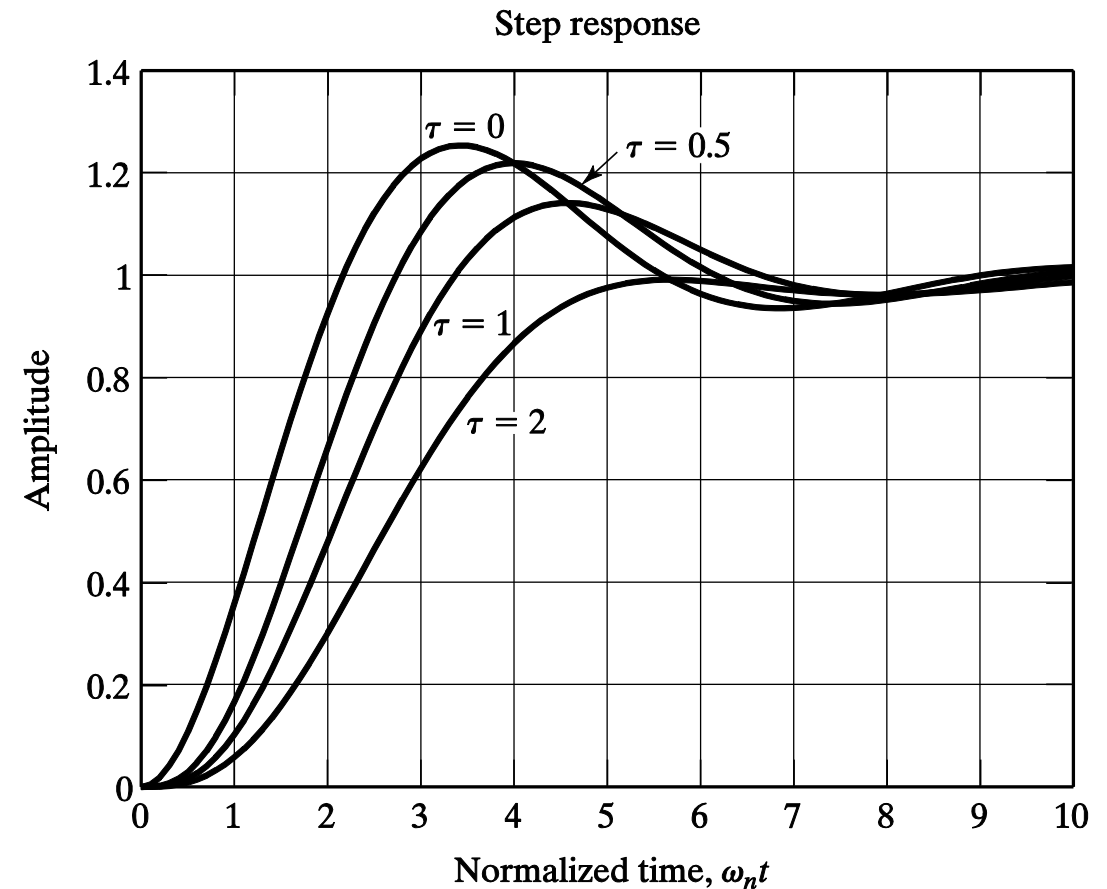
- Consider the case when a **pole** is added to a **standard second-order underdamped system**.

Additional Pole or Zero

- The response of the system will be **slower** and **smoother** than the response of the standard second-order system.
- **The additional low-pass filtering.**

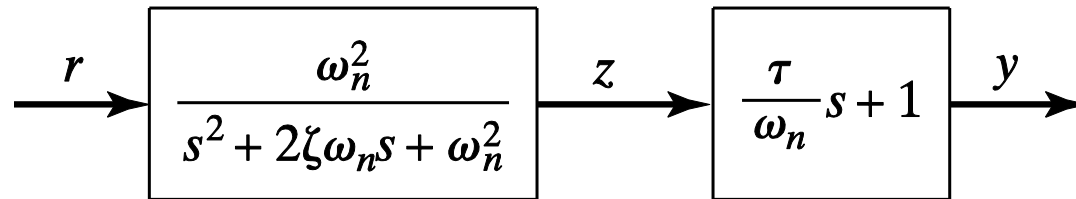


- Effects of a pole on step response with $\zeta = 0.4$.



Additional Pole or Zero

- Consider the case when a **zero** is added to a **standard second-order underdamped system**.



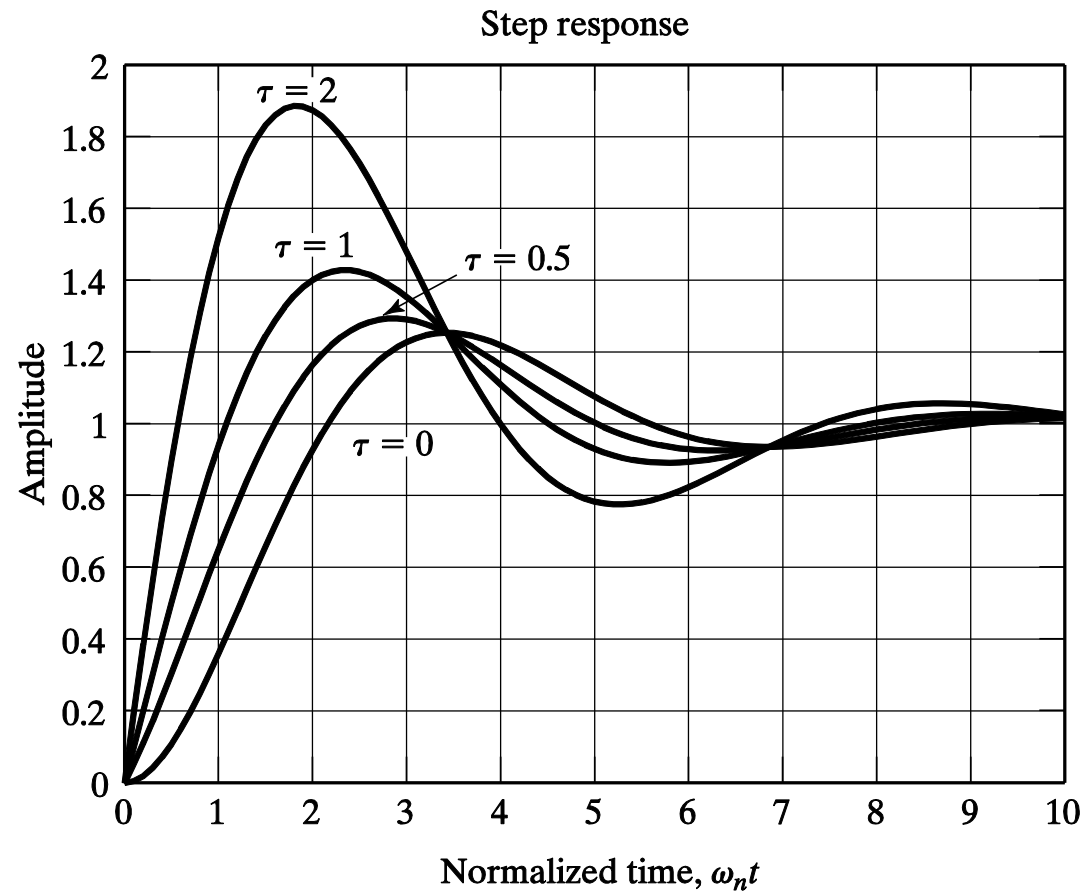
- Assume the zero added is a **minimum phase** zero.
- It is easy to see that

$$y(t) = z(t) + \frac{\tau}{\omega_n} \dot{z}(t).$$

Since $\dot{z}(t) > 0$ for $t < t_p$, the system will have a **shorter rise time**, a **shorter peak time**, and a **larger overshoot**.

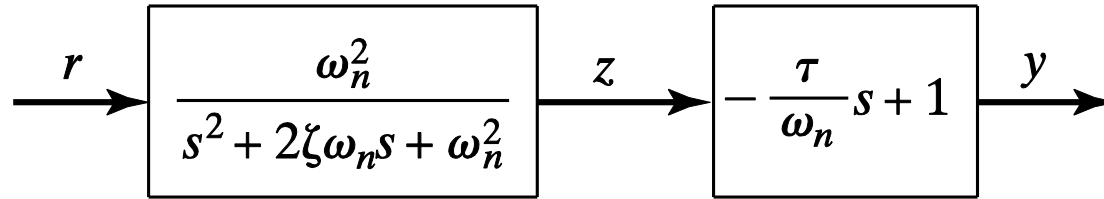
Additional Pole or Zero

- Effects of a zero on step response with $\zeta = 0.4$



Additional Pole or Zero

- Assume the zero added is a **nonminimum phase** zero.



- Then we have

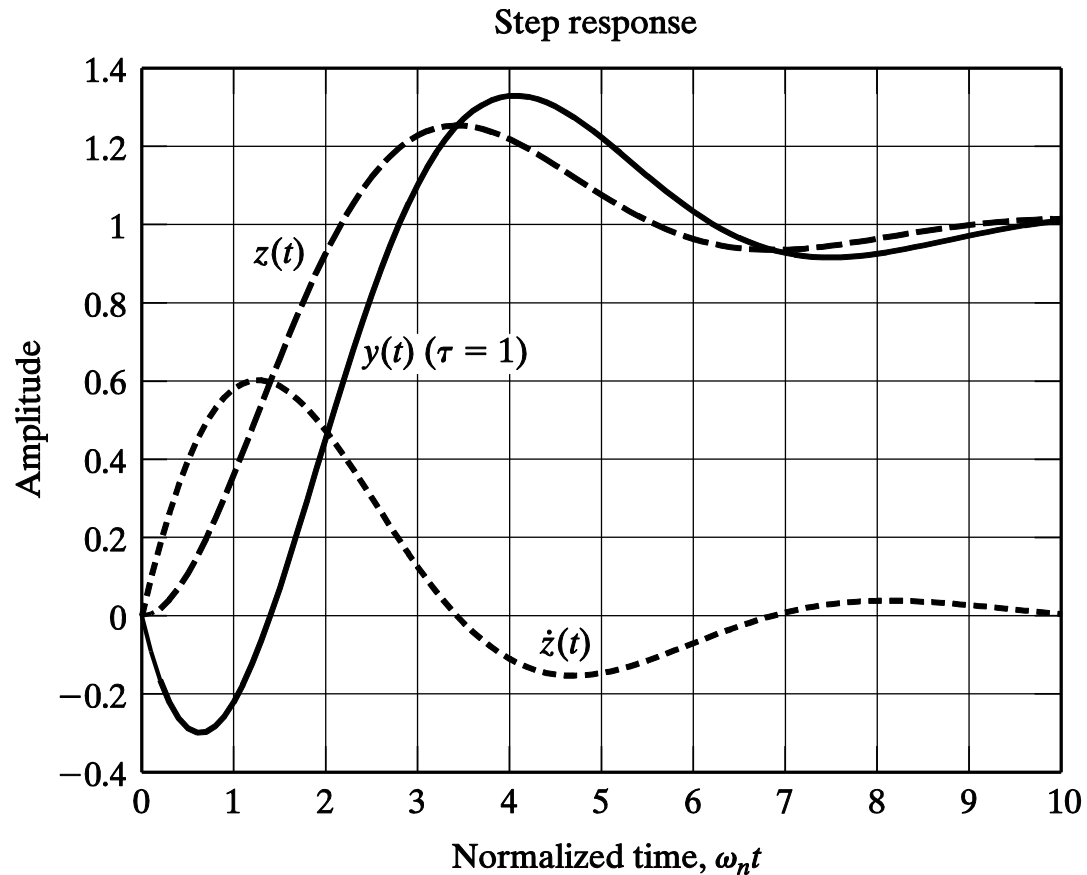
$$y(t) = z(t) - \frac{\tau}{\omega_n} \dot{z}(t).$$

Since $\dot{z}(t) > 0$ for $t < t_p$, the response shows the **undershoot** behavior.

Undesirable!!

Additional Pole or Zero

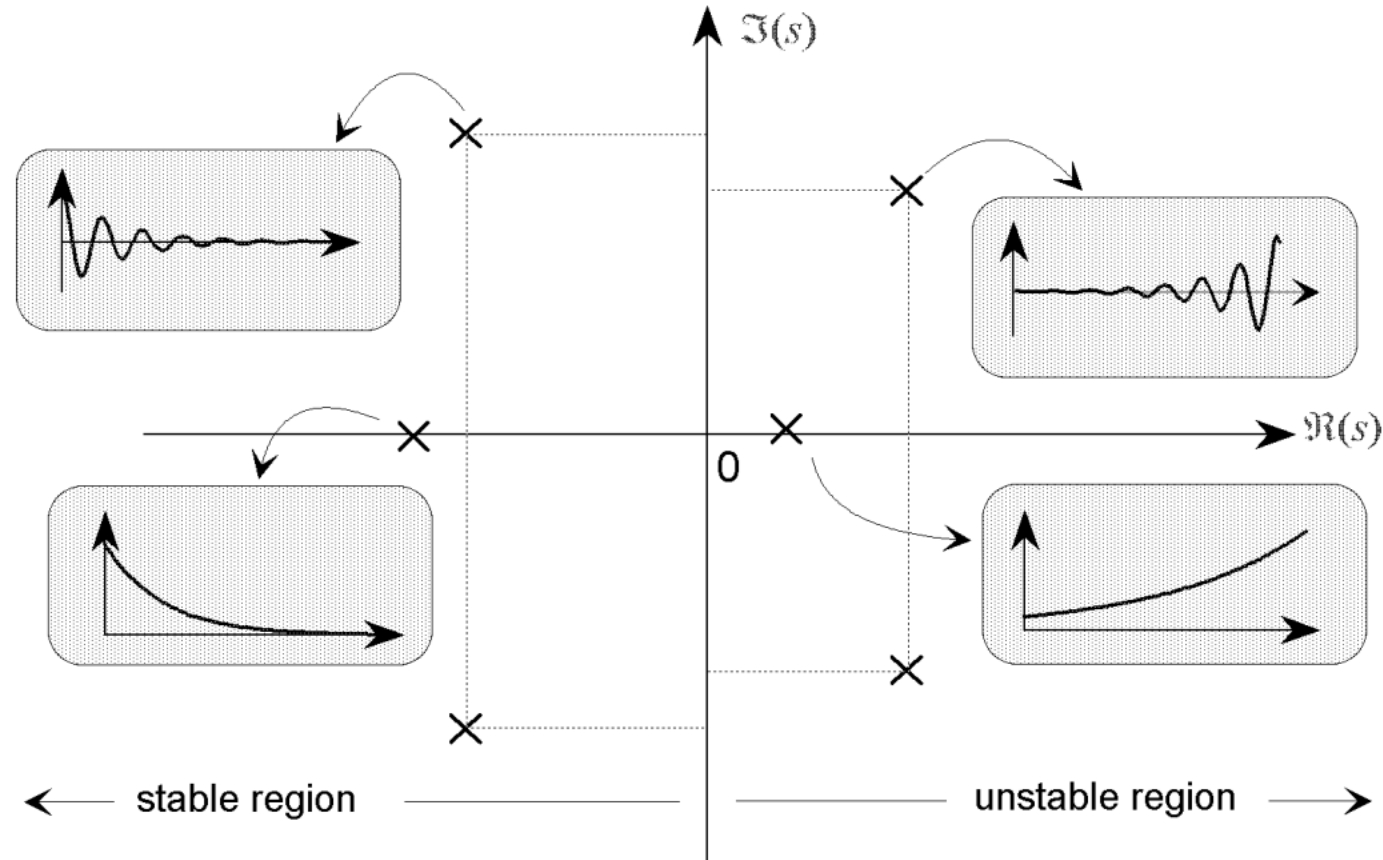
- Effects of a **nonminimum phase zero** on step response with $\zeta = 0.4$



Low-order Approximations

- **Hard** to find **analytic formulas** to evaluate the performance of **high-order systems**.
- Those poles and zeros **far from the imaginary axis** will have **little impact** on the settling time and other performance specifications.
- Good **low-order approximations** of a high-order system can be found by appropriately neglecting the **less significant** poles and zeros.

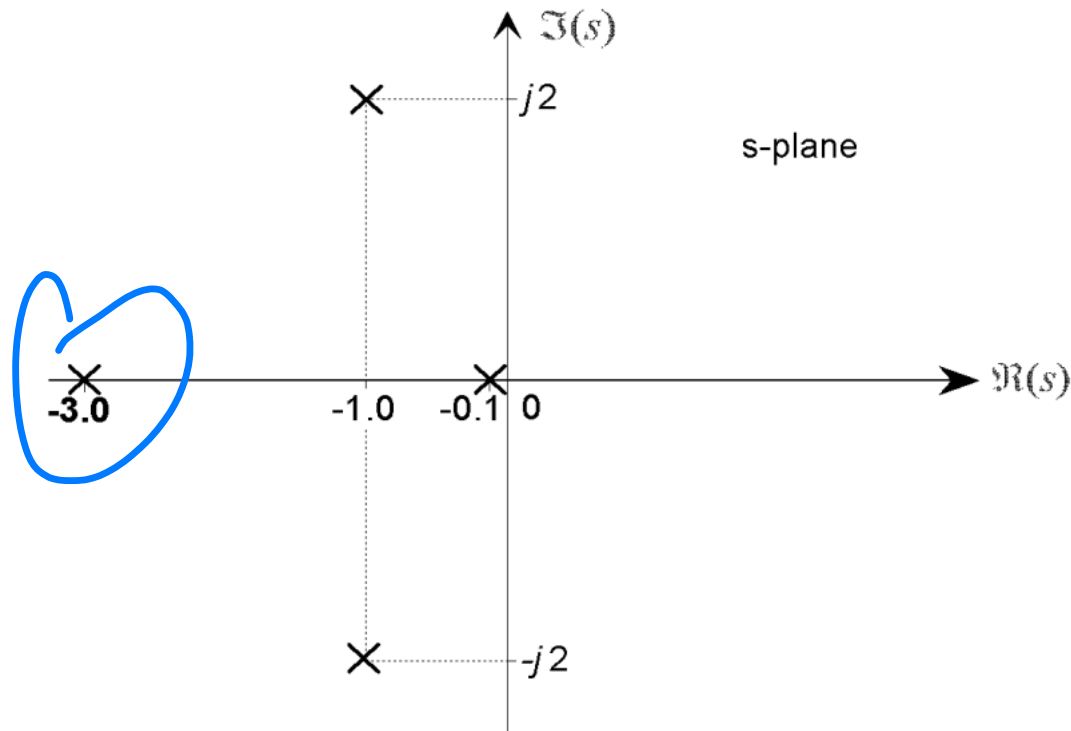
Low-order Approximations



The specification of the form of components of the homogeneous response from the system pole locations on the pole-zero plot.

Low-order Approximations

What's the expected form of the response of a system with a pole-zero plot shown in the following Figure?



Pole-zero plot of a fourth-order system with two real and two complex conjugate poles.

Answer: The system has four poles and no zeros. The two real poles correspond to decaying exponential terms $C_1 e^{-3t}$ and $C_2 e^{-0.1t}$, and the complex conjugate pole pair introduce an oscillatory component $A e^{-t} \sin(2t + \phi)$, so that the total homogeneous response is:

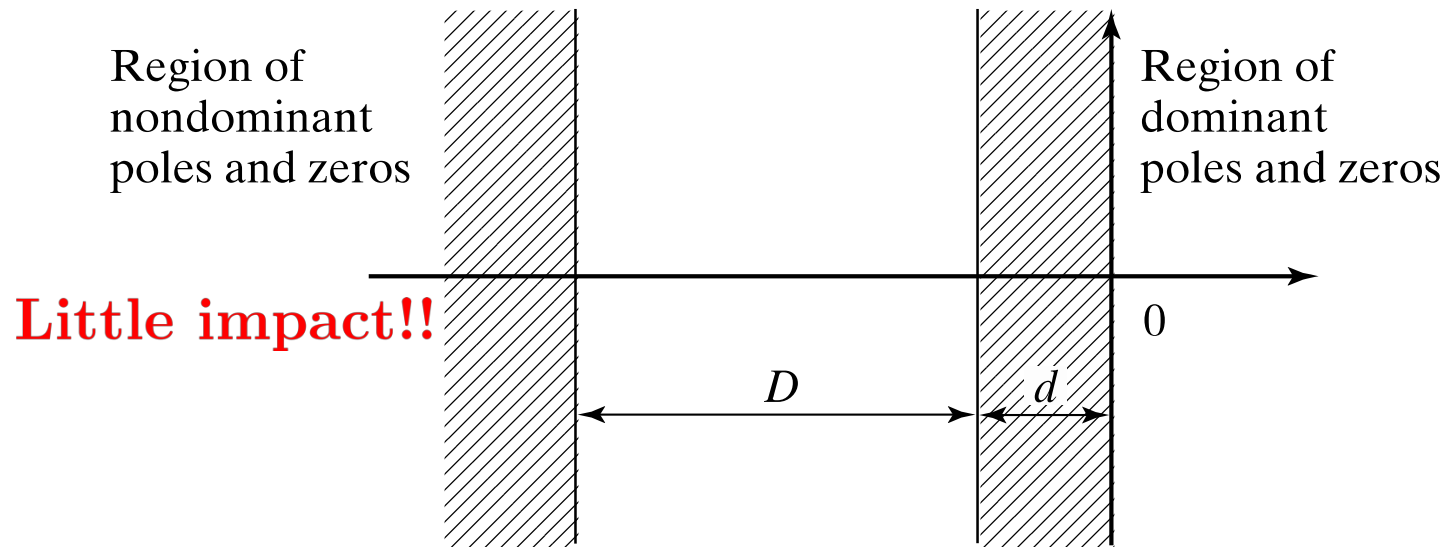
$$y_h(t) = C_1 e^{-3t} + C_2 e^{-0.1t} + A e^{-t} \sin(2t + \phi)$$

The following general observations may be made:

- The term e^{-3t} decays rapidly.
- The term $e^{-0.1t}$ decays slowly. It is therefore the dominant long term response component in the overall homogeneous response.

Dominant Poles and Zeros

- If $D/d > 5 \sim 10$,
 - call those poles (zeros) that are close to the imaginary axis and with **no zeros (poles) close by** as **dominant poles (zeros)**.
 - call those poles (zeros) far from the imaginary axis as **nondominant poles (zeros)**.



Much closer to the imaginary axis!!

Dominant Poles and Zeros

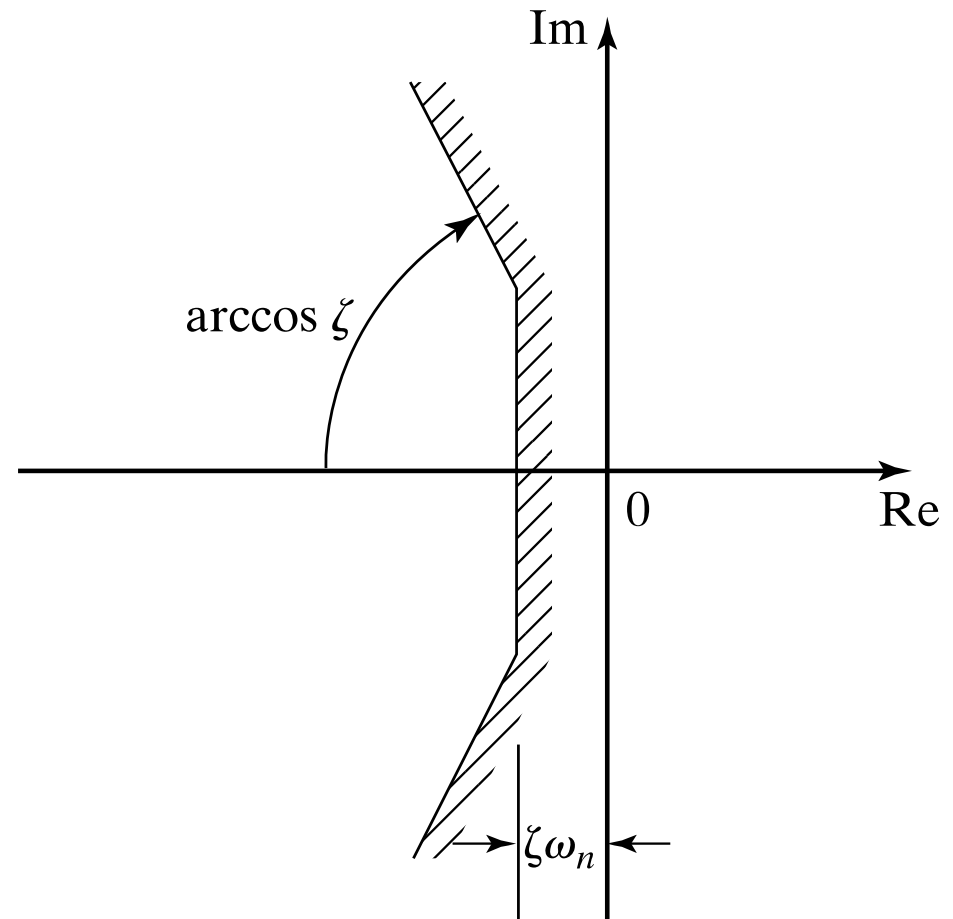
- If the poles are dominated by **a pair of complex poles**, then the system can be approximated by a **second-order** system.
- Establish the **relationship** between
 - the performance specifications
 - the pole locations
- Given the desired t_s and the PO requirements, estimate the **approximate locations** of the dominant closed-loop poles from

$$t_s = \frac{4}{\zeta\omega_n}, \quad \text{PO} = e^{-\zeta\pi/\sqrt{1-\zeta^2}} \times 100\%,$$

$$\zeta\omega_n \geq \frac{4}{t_s}, \quad \zeta \geq \frac{-\ln(\text{PO})}{\sqrt{\pi^2 + (\ln \text{PO})^2}}.$$

The Desired Dominant Pole Region

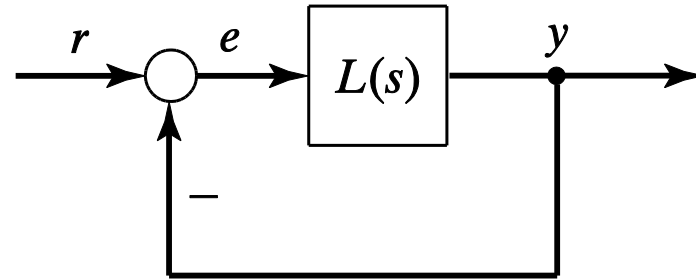
- Place the more **dominant poles** in the **desired regions**.
- Place the less dominant poles far away.



Dominant Poles and Zeros

- **Example** A unity feedback system with loop transfer function

$$L(s) = \frac{10(s + 10)}{s(s + 3)(s + 5)}.$$



- The closed-loop transfer function from $r(t)$ to $y(t)$ is given by

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{10(s + 10)}{(s + 6.5182) [(s + 0.7409)^2 + 3.8461^2]}.$$

Dominant poles!!

Dominant Poles and Zeros

$$T(s) = \frac{L(s)}{1 + L(s)} = \frac{10(s + 10) \text{ Far away.}}{(s + 6.5182) [(s + 0.7409)^2 + 3.8461^2] \text{ Far away.}}$$

- The system performance can be estimated on the basis of the pair of poles.

$$T(s) \approx \frac{10 \times 10}{6.5182 [(s + 0.7409)^2 + 3.8461^2]} = \frac{3.9168^2}{s^2 + 2 \times 0.189 \times 3.9168s + 3.9168^2}.$$

- Natural frequency $\omega_n = 3.92$
- Damping ratio $\zeta = 0.189$

- Thus it is expected that the step response will have

$$t_s = \frac{4}{\zeta \omega_n} = 5.399 \text{ (2\% tolerance), and PO} = e^{-\zeta \pi / \sqrt{1 - \zeta^2}} \times 100\% = 54.63\%.$$

Comparison

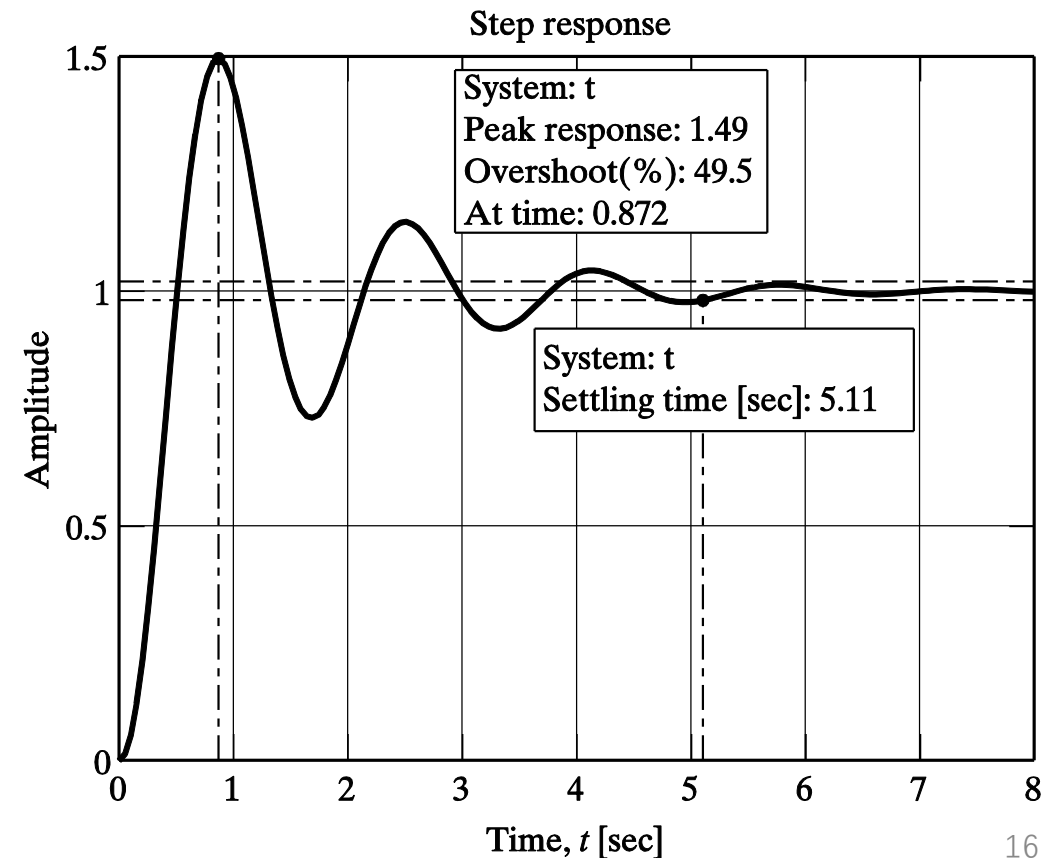
- The **exact** t_s and PO are obtained from numerical simulation:

```
>> L=tf(10*[1,10],[1,8,15,0]);    T=feedback(L,1);
```

```
>> ltiview('step',T)
```

- The **expected** t_s and PO

$$t_s = 5.399,$$
$$PO = 54.63\%.$$



Review

- How the poles and zeros affect the standard system response?
- How the additional poles and zeros affect the system?
- Why can we use a low-order system to approximate a high-order system?
- How to determine the dominant poles and zeros?